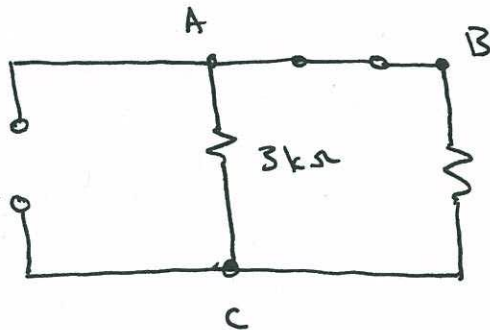


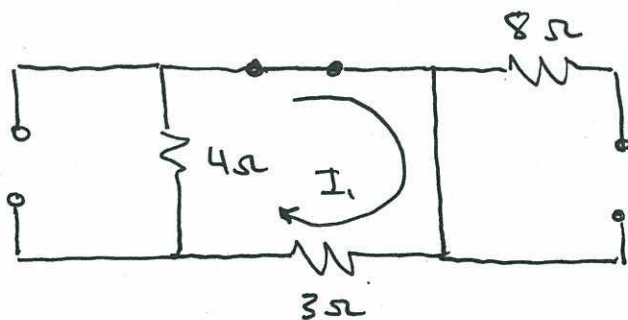
Exercises
Chapter 3.1

1. Kill sources to find independent nodes:

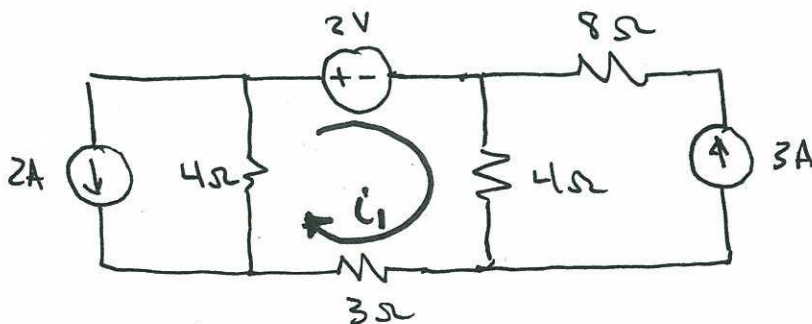


Nodes A & B are dependent. Their voltage difference is set by the 3V source.

2. Kill sources to find meshes:

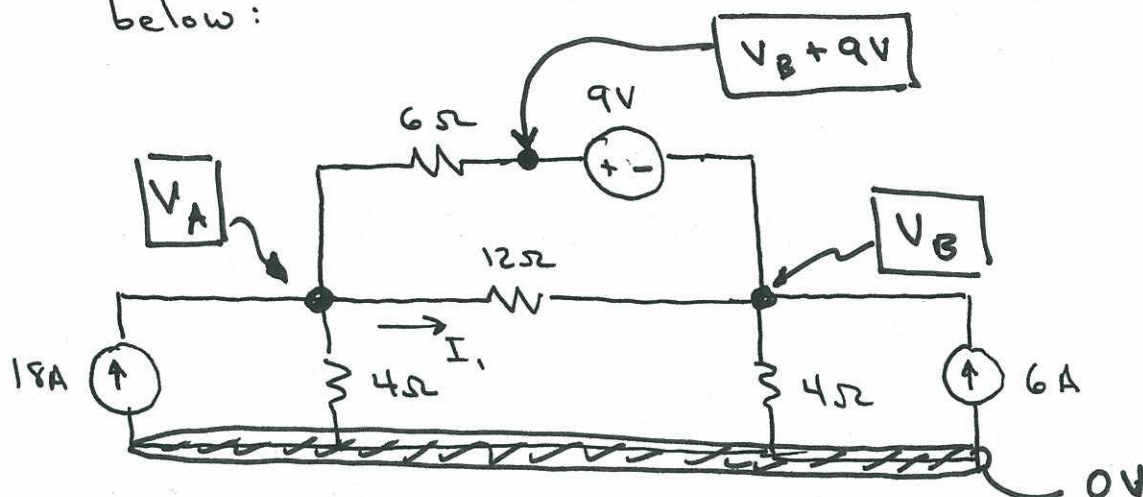


only one enclosed area, so only one mesh.



Exercises
Chapter 3.2

1. Node voltages & dependent voltages shown below:



$$\text{KCL at A: } -18\text{A} + \frac{V_A - 0\text{V}}{4\Omega} + \frac{V_A - V_B}{12\Omega} + \frac{V_A - (V_B + 9\text{V})}{6\Omega} = 0$$

$$\text{KCL at B: } -6\text{A} + \frac{V_B - 0\text{V}}{4\Omega} + \frac{V_B - V_A}{12\Omega} + \frac{(V_B + 9\text{V}) - V_A}{6\Omega} = 0$$

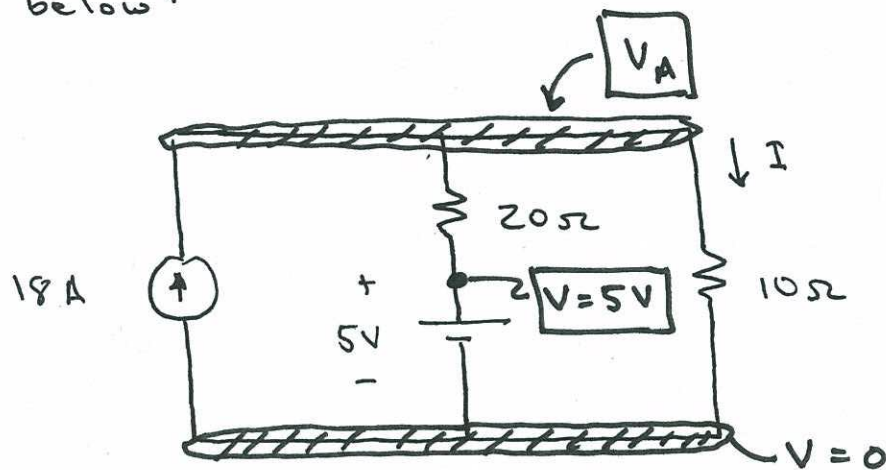
The above are two equations in two unknowns.

I_1 can be found from V_A & V_B :

$$I_1 = \frac{V_A - V_B}{12\Omega}$$

Exercises
Chapter 3.2

2. Node voltages, dependent voltages shown below:



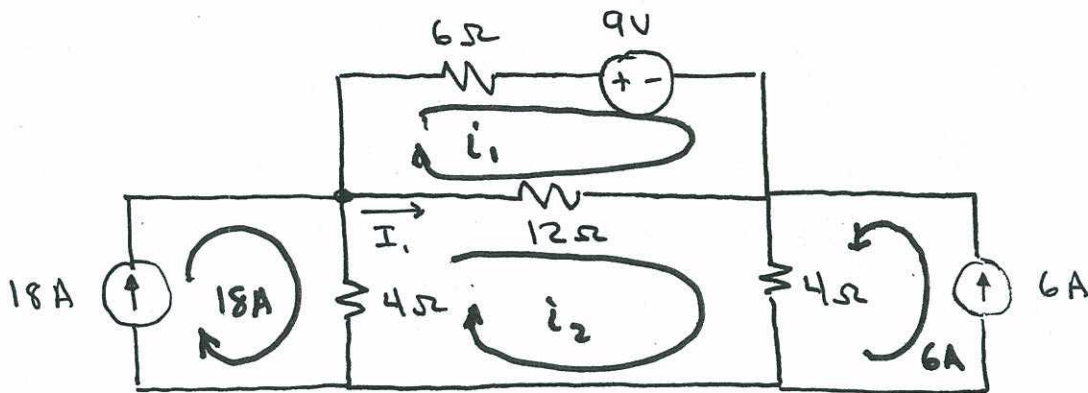
KCL at A:

$$-18A + \frac{V_A - 5V}{20\Omega} + \frac{V_A - 0V}{10\Omega} = 0 \Rightarrow V_A = \frac{365}{3} V$$

$$I = \frac{V_A}{10\Omega} \Rightarrow \underline{\underline{I = \frac{365}{30} A}}$$

Exercises
chapter 3.3

1. Mesh loops, constrained loops shown below:



KVL, Loop i_1 :

$$6\Omega (i_1) + 9V + 12\Omega (i_1 - i_2) = 0 \quad (1)$$

KVL, Loop i_2 :

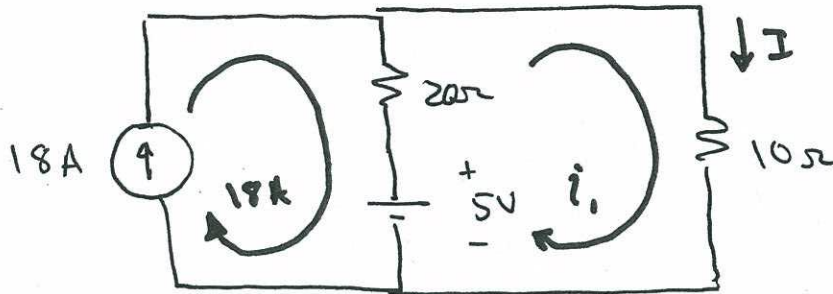
$$4\Omega (i_2 - 18A) + 12\Omega (i_2 - i_1) + 4\Omega (i_2 + 6A) = 0 \quad (2)$$

$$I_1 = i_2 - i_1 \quad (3)$$

3 equations, 3 unknowns!

Exercises
chapter 3.4

2. Mesh i_1 constrained
loops shown below:



KVL, Loop i_1 :

$$-5V + 20\Omega (i_1 - 18A) + 10\Omega (i_1) = 0$$

$$i_1 = \frac{365}{30} A$$

$$I = i_1 \Rightarrow \underline{\underline{I = \frac{365}{30} A}}$$